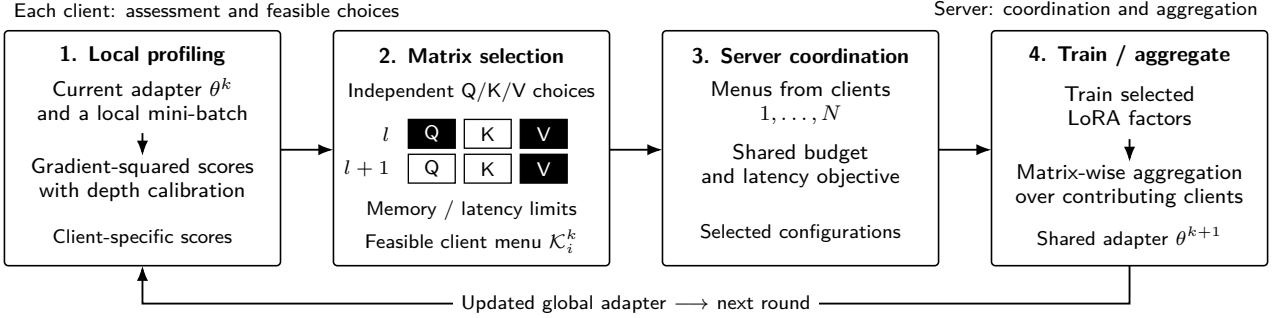

Each client: assessment and feasible choices
Server: coordination and aggregation
1. Local profiling
Current adapter θ^k
and a local mini-batch
Gradient-squared scores
with depth calibration
Client-specific scores
2. Matrix selection
Independent Q/K/V choices
l
Q
K
V
$l+1$
Q
K
V
Memory / latency limits
Feasible client menu $\mathcal{K}_i^k$
3. Server coordination
Menus from clients
$1, \ldots, N$
Shared budget
and latency objective
Selected configurations
4. Train / aggregate
Train selected
LoRA factors
Matrix-wise aggregation
over contributing clients
Shared adapter θ^{k+1}
Updated global adapter ⟶ next round